

Day 2 Report – Data Cleaning & Validation

Mutual Fund Analytics Project

Prepared By

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Executive Summary

The primary objective of Day 2 was to transform raw mutual fund datasets into clean, structured, and analysis-ready data. Data quality is a critical component of any analytics project, as inaccurate or inconsistent data can lead to misleading insights and poor business decisions.

During this phase, multiple datasets were reviewed, validated, and standardized using Python and Pandas. The activities focused on identifying data quality issues, correcting inconsistencies, validating business rules, and preparing the datasets for Exploratory Data Analysis (EDA).

The successful completion of this stage ensured that all subsequent analyses would be performed on reliable and accurate data.

Project Objective

The objectives of Day 2 were:

- Clean and validate raw mutual fund datasets.
- Standardize data formats and structures.
- Remove duplicate and inconsistent records.
- Validate critical business fields.
- Calculate derived metrics required for analysis.
- Prepare datasets for Exploratory Data Analysis (EDA).

Datasets Processed

The following datasets were selected for cleaning and validation:

Dataset	Purpose
02_nav_history.csv	Historical NAV trend analysis
07_scheme_performance.csv	Scheme performance evaluation
08_investor_transactions.csv	Investor behavior and transaction analysis

Data Cleaning Methodology

A structured data quality framework was followed to ensure consistency across all datasets.

The methodology included:

1. Data Profiling
 2. Missing Value Assessment
 3. Duplicate Record Detection
 4. Data Type Validation
 5. Business Rule Validation
 6. Data Standardization
 7. Derived Metric Creation
 8. Final Quality Verification
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NAV History Dataset Cleaning

The NAV History dataset contains historical Net Asset Value records for multiple mutual fund schemes.

Activities Performed

Date Standardization

The date field was converted into a proper datetime format to support time-series analysis and trend identification.

Record Sorting

Records were sorted by:

- AMFI Code
- Date

This ensured chronological consistency for each scheme.

Duplicate Removal

Duplicate NAV records were identified and removed to prevent distortion in trend calculations.

Missing Value Assessment

The dataset was reviewed for null or missing values in critical fields.

NAV Validation

Business validation rules were applied:

- NAV must be greater than zero.
- Invalid or suspicious values were flagged.

Daily Return Calculation

Daily percentage returns were calculated to support:

- Performance analysis
- Correlation analysis
- Volatility assessment

Outcome

The NAV dataset was successfully transformed into a clean and analysis-ready time-series dataset.

Investor Transactions Dataset Cleaning

The Investor Transactions dataset contains information related to purchases, SIP investments, and redemptions.

Activities Performed

Transaction Type Standardization

Transaction categories were standardized into consistent formats such as:

- SIP
- Lumpsum
- Redemption

Amount Validation

Transaction amounts were reviewed to ensure:

- Positive values only
- No invalid entries

Date Validation

Transaction dates were standardized to maintain consistency across records.

KYC Status Verification

KYC values were reviewed to ensure they matched expected business categories.

Demographic Field Validation

The following attributes were reviewed:

- Age Group

- Gender
- State
- City
- City Tier

Outcome

The dataset was standardized and prepared for demographic, geographic, and behavioral analysis.

Scheme Performance Dataset Cleaning

The Scheme Performance dataset contains return and performance-related metrics.

Activities Performed

Numeric Conversion

Performance columns were converted into numeric data types.

Return Validation

All return metrics were validated for consistency and correctness.

Anomaly Detection

Potential outliers and suspicious values were reviewed and flagged for investigation.

Data Consistency Checks

Cross-field consistency checks were performed to ensure reliability.

Outcome

The dataset was successfully prepared for performance comparison and benchmarking activities.

Data Quality Assessment

The following quality checks were performed across all datasets:

Quality Check	Status
Missing Value Analysis	Completed
Duplicate Detection	Completed
Data Type Validation	Completed

Quality Check	Status
Date Standardization	Completed
Business Rule Validation	Completed
Numeric Field Validation	Completed
Category Standardization	Completed

Key Achievements

During Day 2, the following milestones were achieved:

- Successfully cleaned historical NAV records.
 - Standardized investor transaction data.
 - Validated mutual fund performance metrics.
 - Removed duplicate records.
 - Ensured data consistency across datasets.
 - Generated daily return calculations.
 - Created analysis-ready datasets for the EDA phase.
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Challenges Encountered

Several challenges were addressed during the cleaning process:

Inconsistent Date Formats

Different datasets contained varying date formats that required standardization.

Transaction Category Variations

Transaction type values needed normalization to ensure consistency.

Large Historical Dataset

The NAV dataset contained a substantial volume of historical records requiring validation and sorting.

Business Rule Verification

Additional validation was required to ensure values complied with expected financial rules.

Business Impact

High-quality data significantly improves the reliability of analytical results.

The cleaning and validation process completed on Day 2 provides:

- Improved analytical accuracy
 - Reliable performance measurements
 - Consistent reporting metrics
 - Better decision-making support
 - Stronger foundation for visualization and dashboard development
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Conclusion

Day 2 successfully achieved its objective of transforming raw mutual fund datasets into structured, validated, and analysis-ready data. Through comprehensive cleaning, validation, and standardization activities, data quality was significantly improved, ensuring that subsequent analyses would be based on accurate and reliable information.

The cleaned datasets now provide a robust foundation for Exploratory Data Analysis (EDA), trend identification, visualization development, and business insight generation in the next phase of the project.

Deliverables

- Cleaned NAV History Dataset
 - Cleaned Investor Transactions Dataset
 - Cleaned Scheme Performance Dataset
 - Daily Return Calculations
 - Data Validation Documentation
 - Analysis-Ready Datasets
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Git Commit

Day 2: Data cleaning and validation completed

Author

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